

From benchmark performance to auditable storage decisions

A prospective evidence-transfer pathway. Each gate adds a distinct evidence artifact; later evidence does not erase an earlier gap.

Qualification stage	Evidence artifact produced	Stop if absent	Next admissible claim
S1 Provenance-ready benchmark Gate 1 of 7	Versioned benchmark and provenance dossier	Provenance, split independence, label origin or use-domain limits cannot be reconstructed.	The workflow is reproducible within a defined benchmark domain.
S2 Physically diagnosed model Gate 2 of 7	Decision-variable physical diagnostic report	Only aggregate predictive error is reported, or the physical relation controlling the decision is neither encoded nor quantitatively checked.	The model is physically diagnosed for the evaluated process and regime.
S3 Uncertainty-qualified workflow Gate 3 of 7	Calibration and discrepancy-propagation dossier	An ensemble, posterior or interval is presented without a reliability check, or a surrogate is treated as exact in downstream analysis.	Uncertainty is reliability-checked and propagated for the evaluated workflow, to the extent explicitly demonstrated.
S4 Shift-tested workflow Gate 4 of 7	Geological and operational stress-test matrix	Evaluation remains a random same-distribution split, or the claimed shift is not defined relative to development data.	The workflow has been tested under the named geological or operational shifts.
S5 Field-evaluated update Gate 5 of 7	Field-evaluation and monitoring-update dossier	A named site, field-derived input or field-inspired simulator is used without comparing the ML output with field or controlled evidence.	The workflow has single-site or controlled-experiment evidence in the stated setting.
S6 Reviewable decision support Gate 6 of 7	Decision protocol and human-review record	The model output is not linked to a named action, consequence-aware metric, safe fallback or accountable reviewer.	The workflow has been evaluated as decision support for the stated case and review process.
S7 Auditable evidence system Gate 7 of 7	MRV and stewardship audit package	Data, model, update, decision or corrective-action provenance cannot be reconstructed, or review responsibility is undefined.	The workflow is auditable for MRV or stewardship within the demonstrated scope.

Claim boundary: the pathway is review-derived author synthesis, not an observed maturity ladder; universal standard or regulatory protocol. Unknown, unclear and not reported remain missing evidence rather than zero.